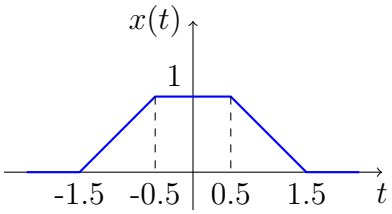
EE1101 Signals and Systems JAN—MAY 2020
Tutorial 8

`Notation:` In this tutorial, the CTFT of $x(t)$ is denoted by $X(j\omega)$. Another commonly used notation for CTFT is $X(\omega)$. Hence, $X(j\omega)$ and $X(\omega)$ represent the same spectrum.

1. Given two signals $f_1(t) = 2\,\text{rect}(t/4)$ and $f_2(t) = \text{rect}(t/2)$
   (a) Sketch $g(t) = f_1(t) \star f_2(t)$
   (b) Find $G(j\omega)$, the Fourier transform of $g(t)$, directly using the analysis equation.
   (c) Find $G(j\omega)$ using the convolution property of Fourier transform.
   (d) Sketch the magnitude and phase spectrum of $G(j\omega)$

$$\textit{Note :}\ \ \text{rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1 & ,|t| \leq \frac{\tau}{2} \\ 0 & ,|t| > \frac{\tau}{2} \end{cases}$$

2. Using relevant properties, find the Fourier transform $X(j\omega)$ of the signal $x(t)$ shown below.



   `Hint:` Results from problem 1 may be useful.

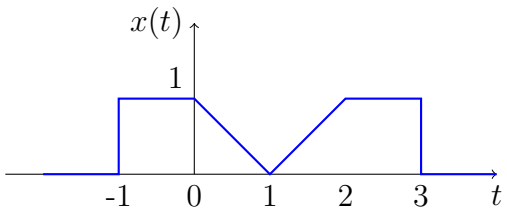
3. Determine the signal $x(t)$ corresponding to the following Fourier transforms:
   (a) $X(j\omega) = \dfrac{2\sin[3(\omega - 2\pi)]}{(\omega - 2\pi)}$
   (b) $X(j\omega) = \cos\left(4\omega + \dfrac{\pi}{3}\right)$

4. (a) Find the Fourier transform of the signal $x(t) = e^{-a|t|}$, $a > 0$.
   (b) Using the answer to previous part, find the energy of the signal $x(t) = \frac{1}{t^2+1}$.

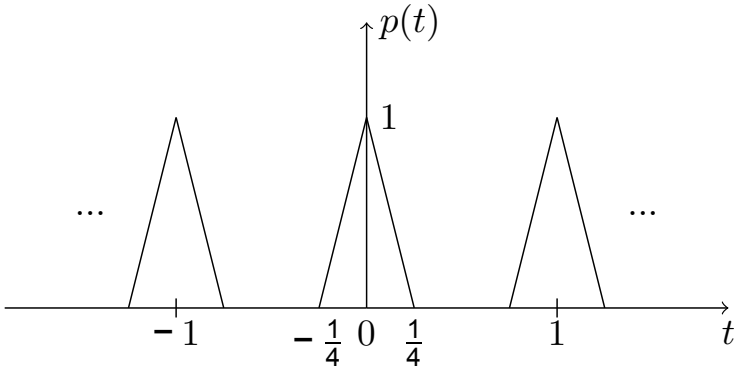
5. Let $X(j\omega)$ be the Fourier transform of the signal $x(t)$ shown below. Do the following computations without explicitly finding $X(j\omega)$.
   (a) Find $X(0)$
   (b) Evaluate $\int_{-\infty}^{\infty} X(j\omega)d\omega$
   (c) Evaluate $\int_{-\infty}^{\infty} X(j\omega)e^{j\omega}d\omega$

(d) Evaluate $\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$

(e) Sketch the inverse Fourier transform of $\texttt{Real}\{X(j\omega)\}$.



6. Let $p(t)$ be periodic triangular pulse train shown below.



(a) Calculate P_n, the exponential Fourier series coefficients of $p(t)$ and sketch P_n vs. n.

(b) Calculate $P(j\omega)$, the Fourier transform of $p(t)$. Sketch $P(j\omega)$ vs. ω.

(c) Let $x(t)$ be an aperiodic signal having Fourier transform $X(j\omega)$. Define $y(t) = p(t) \cdot x(t)$. Find an expression for $Y(j\omega)$, the Fourier transform of $y(t)$.

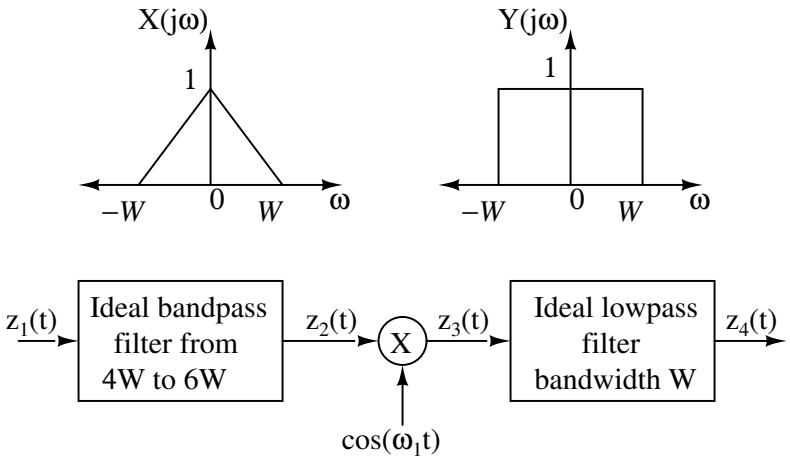
(d) For the previous part, if $x(t) = \text{sinc}(t)$, find and sketch $Y(j\omega)$.

7. Let $x(t) = e^{-at}u(t)$ be given as an input to an LTI system with frequency response $H(j\omega) = \frac{1}{a+j\omega}$. Let $y(t)$ denote the output.

(a) Find the Fourier transform $Y(j\omega)$ of the output $y(t)$.

(b) Using relevant properties of Fourier transform, find the expression for the output signal $y(t)$.

8. $x(t)$ and $y(t)$ have Fourier transforms as shown below. Sketch the Fourier transform of the various signals $z_i(t)$ for $i = 1, 2, 3, 4$ in the system shown below given that $z_1(t) = x(t)\cos(\omega_1 t) + y(t)\cos(\omega_2 t)$. Determine $z_4(t)$ in terms of $x(t)$ and $y(t)$? Assume that $\omega_1 = \omega_2 - 2W = 5W$.


X(jω)
1
−W
0
W
ω
Y(jω)
1
−W
0
W
ω
z1(t)
Ideal bandpass filter from 4W to 6W
z2(t)
X
z3(t)
Ideal lowpass filter bandwidth W
z4(t)
cos(ω1t)